



BAUST CSE FEST 2026 Hackathon Challenge

Problem Statement: Smart Home Service Automation

People regularly need services such as appliance repair, plumbing, electrical work, cleaning, pest control, moving, car servicing, and home maintenance.

Today, arranging these services often involves multiple manual steps:

- Finding an appropriate service provider
- Checking availability
- Comparing providers
- Choosing a suitable time
- Sharing the service location and requirements
- Following up on the request
- Tracking whether the service has been completed

Your challenge is to build a **smart service management system that automates this process**.

The user should be able to select a service, provide the necessary details, choose their preferred time, and submit a request. The system should then automatically find suitable service providers based on factors such as **service type, location, availability, rating, price, and urgency**.

The system should minimize the amount of manual coordination required from both the customer and the service provider.

Example Scenario

A customer needs an AC repair.

They select:

Service: AC Repair
Location: Dhanmondi
Preferred Date: Saturday
Preferred Time: 4:00 PM – 6:00 PM
Urgency: Normal

The system checks available technicians and automatically recommends suitable options.

For example:

Recommended Technician
Rahim Electronics
• 4.8 rating
• 2.3 km away
• Estimated service charge: ₳1,000
• Available: 4:30 PM

The customer confirms the request.

The technician receives the job and can:

Accept → On the Way → In Progress → Completed

The customer can track the status throughout the process.

Minimum Requirements

Your MVP should include:

1. Service Selection

Allow customers to select from several service categories, such as:

- Appliance & Gadget Repair
- Plumbing
- Electrical
- Cleaning & Pest Control
- Home Maintenance
- Moving & Shifting
- Car Care & Repair
- Personal Care

2. Service Request

The customer should be able to provide relevant information such as:

- Service type
- Location
- Preferred date/time
- Urgency
- Basic problem details
- Optional image
- Contact information

3. Smart Provider Matching

Create a small mock database of service providers.

The system should recommend providers based on multiple factors, for example:

Match Score = Availability + Distance + Rating + Price + Service Expertise

Teams are free to design their own matching algorithm.

4. Scheduling

The system should prevent double-booking and show available time slots.

A provider should not be offered to two customers for the same time slot.

5. Request Tracking

The customer should be able to see:

Requested → Accepted → On the Way → In Progress → Completed

6. Provider Dashboard

Service providers should be able to:

- View incoming requests
- Accept/reject jobs
- View customer details
- See scheduled jobs
- Update job status

Bonus Challenges

Teams that complete the core system can implement additional automation such as:

- Automatic provider assignment
- Emergency/urgent job prioritization
- Automatic rescheduling when a provider becomes unavailable
- Route/distance-based provider selection
- Price comparison
- Customer notifications
- Provider workload balancing
- Service history
- Automatic invoice generation
- Customer rating and feedback

Constraints

This is an **8-hour hackathon**.

Teams are **not expected to build a production-ready marketplace**.

They may use:

- Mock service providers
- Mock locations
- Mock pricing
- Simulated notifications
- Sample service data

The goal is to demonstrate a **working automation workflow**, not a complete commercial platform.

Core Question

How can technology reduce the manual effort required to request, assign, schedule, and complete everyday services?